

Every year, the moon drifts 1.48 in (3.78 cm) further away from Earth.

Earth's inner core spins at a different speed to the rest of the planet.

More than 300,000 impact craters wider than 0.6 miles (1 km) cover the moon's surface.

The moon

Orbiting Earth every 27 days, the moon is a familiar sight in the night sky. The same side of the moon always faces Earth. The dark side of the moon can only be seen from spacecraft.



Inner core

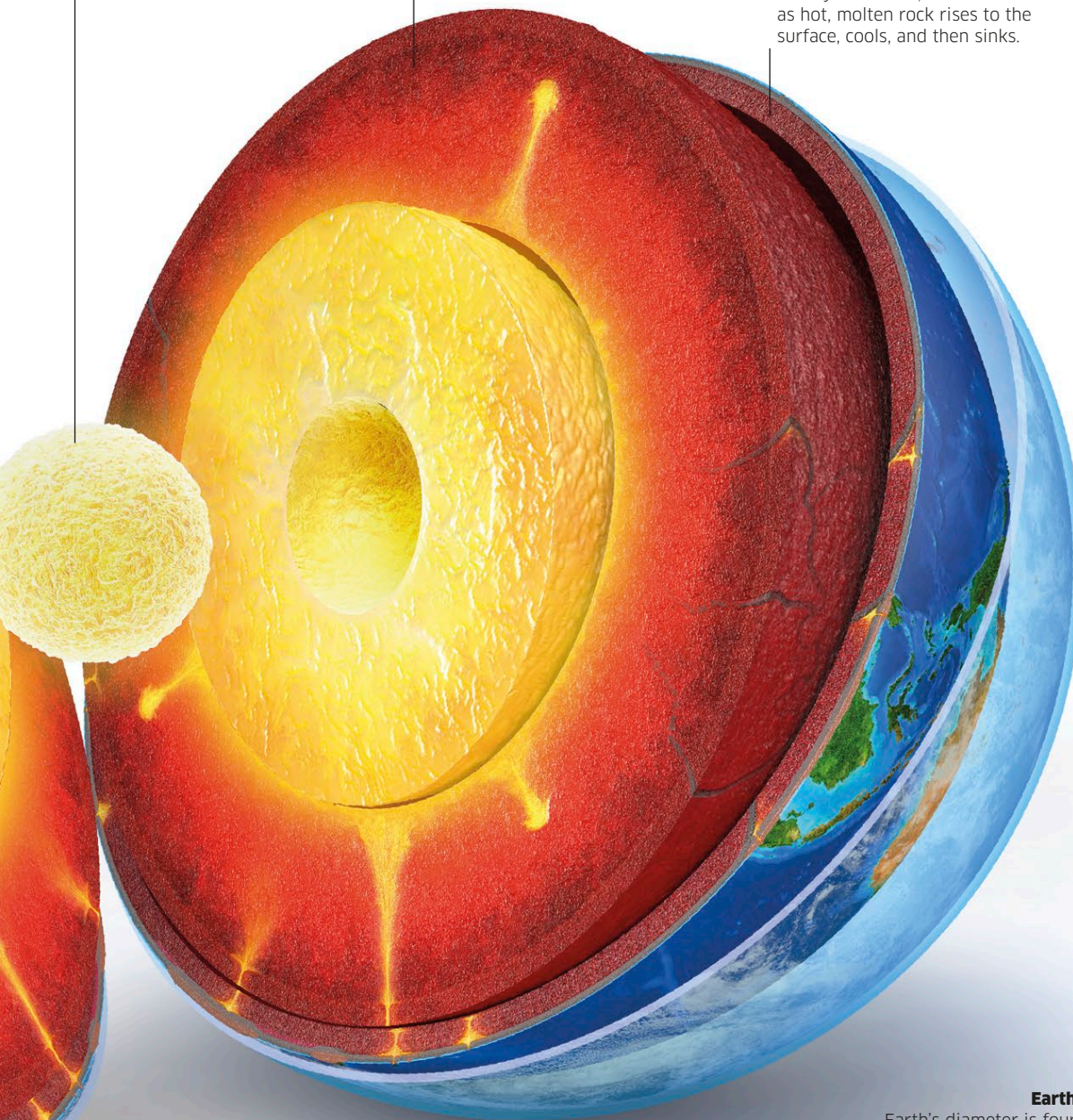
The iron inner core is just over two-thirds of the size of the moon and as hot as the surface of the sun. It is solid because of the immense pressure on it.

Lower mantle

The lower layer of the mantle contains more than half the planet's volume and extends 1,800 miles (2,900 km) below the surface. It is hot and dense.

Upper mantle

The layer extending 255 miles (410 km) below the crust is mostly solid rock, but it moves as hot, molten rock rises to the surface, cools, and then sinks.



Earth to sun

The sun is 93 million miles (150 million km) from Earth. It takes light 8 minutes to travel this distance, known as one astronomical unit (AU).

Earth

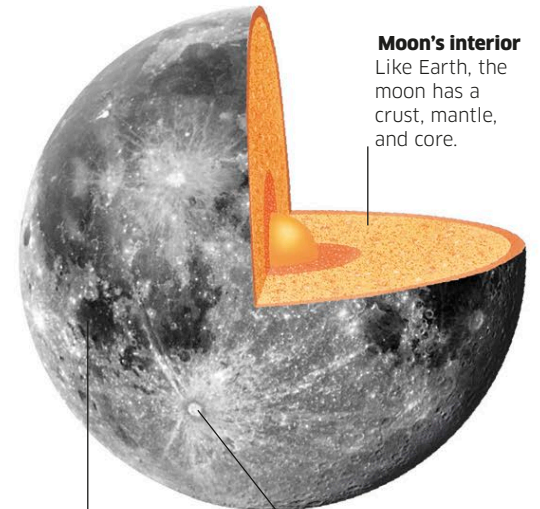
Earth's diameter is four times that of the moon, and our planet weighs 80 times more than its satellite.

Moon

The moon is 239,000 miles (384,000 km) from Earth.

Moon

Our only natural satellite, the moon is almost as old as Earth. It is thought it was made when a flying object the size of Mars crashed into our planet, knocking lots of rock into Earth's orbit. This rock eventually clumped together to form our moon. It is the moon's gravitational pull that is responsible for tides.



Moon's interior

Like Earth, the moon has a crust, mantle, and core.

Lunar maria

Dark, flat areas known as maria, or seas, are in fact huge plains of solidified lava.

Lunar craters

Craters, formed by asteroid impacts 3.5 billion years ago, pockmark the moon.

Lunar cycle

The moon doesn't produce its own light. The sun illuminates exactly half of the moon, and the amount of the illuminated side we see depends upon where the moon is in its orbit around Earth. This gives rise to the phenomenon known as the phases of the moon.



NEW MOON
(0 DAYS)



WAXING
CRESCENT



FIRST QUARTER
(DAY 7)



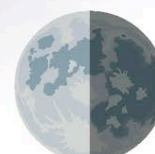
WAXING
GIBBOUS



FULL MOON
(DAY 14)



WANING
GIBBOUS



LAST QUARTER
(DAY 21)



WANING
CRESCENT



NEW MOON
(DAY 28)